

Comprehensive Documentation: Pansharpening Image Fusion using Multi-Scale Wavelet Vision Transformers (HWViT) and NSGA-II

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I. MULTI-OBJECTIVE OPTIMIZATION USING NSGA-II

A. Overview

In this work, we formulate the design of the fusion model as a **multi-objective optimization problem (MOP)**, where multiple conflicting objectives must be optimized simultaneously. Unlike single-objective optimization, no single optimal solution exists; instead, we aim to approximate the **Pareto-optimal set**.

We employ the **Non-dominated Sorting Genetic Algorithm II (NSGA-II)**, a widely used evolutionary algorithm for solving multi-objective problems efficiently.

B. Mathematical Formulation

The general multi-objective optimization problem is defined as:

$$\min_{\mathbf{x} \in \Omega} \mathbf{F}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_M(\mathbf{x})] \quad (1)$$

subject to:

$$\mathbf{x} = (x_1, x_2, \dots, x_n), \quad x_i \in [x_i^{(l)}, x_i^{(u)}] \quad (2)$$

where:

- $\mathbf{x}$ is the decision vector (hyperparameters),
- $\mathbf{F}(\mathbf{x})$ is the objective vector,
- M is the number of objectives,
- Ω is the feasible search space.

C. Decision Variables

In our implementation, the decision vector is:

$$\mathbf{x} = [x_1, x_2, x_3, x_4, x_5] \quad (3)$$

where:

- x_1 : PAN target channels $\in [16, 64]$
- x_2 : MS target channels $\in [16, 64]$
- x_3 : head channel size $\in [4, 16]$
- x_4 : dropout rate $\in [0.01, 0.3]$
- x_5 : learning rate $\in [10^{-4}, 10^{-3}]$

Thus, the search space is:

$$\Omega = [16, 64] \times [16, 64] \times [4, 16] \times [0.01, 0.3] \times [10^{-4}, 10^{-3}] \quad (4)$$

D. Constraint Handling via Parameter Projection

The architecture imposes a structural constraint:

$$\text{channels} \bmod \text{head_channel} = 0 \quad (5)$$

To enforce this, we project raw variables onto the feasible set:

$$x^{\text{proj}} = \max\left(h, \left\lfloor \frac{x}{h} \right\rfloor \cdot h\right) \quad (6)$$

where $h = x_3$ (head channel). This ensures compatibility with multi-head attention reshaping:

$$\text{channels} = (\text{num_heads}) \times (\text{head_channel}) \quad (7)$$

E. Objective Functions

We define a **7-objective minimization problem**:

$$\mathbf{F}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ f_3(\mathbf{x}) \\ f_4(\mathbf{x}) \\ f_5(\mathbf{x}) \\ f_6(\mathbf{x}) \\ f_7(\mathbf{x}) \end{bmatrix} \quad (8)$$

Each objective is derived from evaluation metrics:

$$f_1 = \frac{\text{SAM}}{10} \quad (9)$$

$$f_2 = \frac{\text{ERGAS}}{10} \quad (10)$$

$$f_3 = -\text{CC} \quad (11)$$

$$f_4 = \frac{\text{SD}}{10} \quad (12)$$

$$f_5 = -\frac{\text{SF}}{10} \quad (13)$$

$$f_6 = -\text{SSIM} \quad (14)$$

$$f_7 = \frac{\text{n_params}}{10^6} \quad (15)$$

Interpretation:

- SAM, ERGAS, SD $\rightarrow$ minimized (error metrics)
- CC, SF, SSIM $\rightarrow$ maximized $\Rightarrow$ negated
- Model size $\rightarrow$ minimized

Thus, all objectives are converted into a **pure minimization problem**.

F. Pareto Dominance

A solution $\mathbf{x}^{(1)}$ dominates $\mathbf{x}^{(2)}$ if:

$$\forall i : f_i(\mathbf{x}^{(1)}) \leq f_i(\mathbf{x}^{(2)}) \quad \text{and} \quad \exists j : f_j(\mathbf{x}^{(1)}) < f_j(\mathbf{x}^{(2)}) \quad (16)$$

The goal is to approximate the **Pareto front**:

$$\mathcal{P}^* = \{\mathbf{x} \in \Omega \mid \nexists \mathbf{x}' \prec \mathbf{x}\} \quad (17)$$

G. NSGA-II Algorithm

NSGA-II operates using the following key steps:

1) *1. Initialization*: A population of size N is randomly sampled:

$$P_0 = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \quad (18)$$

In our case:

$$N = \text{pop_size} \quad (19)$$

2) *2. Evaluation*: Each individual is evaluated using:

$$\mathbf{F}(\mathbf{x}_i) = \text{train_and_evaluate}(\mathbf{x}_i) \quad (20)$$

This involves:

- Training the model for a fixed number of epochs
- Evaluating on the test dataset
- Computing all 7 metrics

3) *3. Non-Dominated Sorting*: The population is sorted into fronts:

- F_1 : non-dominated solutions
- F_2 : dominated only by F_1
- etc.

Each solution is assigned a **rank**:

$$\text{rank}(\mathbf{x}) = k \quad \text{if } \mathbf{x} \in F_k \quad (21)$$

4) *4. Crowding Distance*: To preserve diversity, a crowding distance is computed:

$$d_i = \sum_{m=1}^M \frac{f_m^{i+1} - f_m^{i-1}}{f_m^{\max} - f_m^{\min}} \quad (22)$$

This measures how isolated a solution is in objective space.

5) *5. Selection*: Binary tournament selection is applied based on:

- Lower rank preferred
- Higher crowding distance preferred

6) *6. Genetic Operators*: Offspring are generated via:

- Simulated Binary Crossover (SBX)
- Polynomial Mutation

7) *7. Survival (Elitism)*: Parents and offspring are merged:

$$R_t = P_t \cup Q_t \quad (23)$$

Then the next generation is selected:

$$P_{t+1} = \text{best } N \text{ solutions from } R_t \quad (24)$$

based on rank and crowding distance.

H. Algorithm Execution in Code

The optimization is executed as:

$$\text{minimize}(\text{problem}, \text{algorithm}, ('n_gen', G)) \quad (25)$$

where:

- G is the number of generations
- Each generation evaluates N individuals

Total evaluations:

$$N_{\text{eval}} = N \times G \quad (26)$$

I. Checkpointing Mechanism

To ensure robustness, the algorithm state is periodically saved:

$$\mathcal{A}_t \rightarrow \text{disk} \quad (27)$$

This includes:

- Population
- Objective values
- Generation count

If interrupted, optimization resumes from:

$$t = \text{algorithm.n_gen} \quad (28)$$

J. Model Caching via Hashing

Each evaluated configuration is uniquely identified:

$$h(\mathbf{x}) = \text{MD5}(\mathbf{x}) \quad (29)$$

This ensures:

- No duplicate training
- Reproducibility

K. Summary

The proposed NSGA-II framework:

- Optimizes 5 architectural hyperparameters
- Balances 7 competing objectives
- Produces a Pareto front of optimal trade-offs
- Ensures architectural feasibility via projection
- Supports long-running experiments via checkpointing

This formulation enables systematic exploration of the trade-off between reconstruction quality and model complexity in hyperspectral image fusion.

II. NSGA-II BASED HYPERPARAMETER OPTIMIZATION WITH EMBEDDED TRAINING PIPELINE

A. Overview

The proposed framework formulates hyperspectral image fusion as a **multi-objective hyperparameter optimization problem**, where each candidate solution corresponds to a complete training and evaluation cycle of the deep learning model.

Unlike conventional optimization, the objective function is not analytically defined but instead computed via a **computationally expensive training pipeline**. Thus, NSGA-II operates as a **wrapper optimization algorithm**, where:

$$\mathbf{F}(\mathbf{x}) = \text{TrainEval}(\mathbf{x}) \quad (30)$$

Each evaluation involves:

- Dataset loading and preprocessing
- Model instantiation
- Full training over multiple epochs
- Validation-based metric computation

B. Decision Space and Architecture Constraints

Each individual in NSGA-II represents:

$$\mathbf{x} = [x_1, x_2, x_3, x_4, x_5] \quad (31)$$

where:

$$x_1 \rightarrow \text{PAN target channels} \quad (32)$$

$$x_2 \rightarrow \text{MS target channels} \quad (33)$$

$$x_3 \rightarrow \text{head channel size} \quad (34)$$

$$x_4 \rightarrow \text{dropout rate} \quad (35)$$

$$x_5 \rightarrow \text{learning rate} \quad (36)$$

1) *Feasibility Projection*: Due to multi-head attention constraints:

$$\text{channels} = H \times C_h \quad (37)$$

where H is the number of heads and $C_h = x_3$.

Thus, we enforce:

$$x_{\text{proj}} = \max \left(C_h, \left\lfloor \frac{x}{C_h} \right\rfloor C_h \right) \quad (38)$$

applied to both PAN and MS channels. This guarantees valid tensor reshaping:

$$\mathbb{R}^{B \times C \times H \times W} \rightarrow \mathbb{R}^{B \times H \times C_h \times H \times W} \quad (39)$$

C. Embedded Training Pipeline

1) *Dataset Loading and Normalization*: Given dataset $\mathcal{D}$ stored in HDF5 format:

$$\mathcal{D} = \{\mathbf{I}_{\text{pan}}, \mathbf{I}_{\text{ms}}, \mathbf{I}_{\text{ms}}, \mathbf{I}_{\text{gt}}\} \quad (40)$$

each component is normalized:

$$\tilde{\mathbf{I}} = \frac{\mathbf{I}}{r}, \quad r = 2047 \quad (41)$$

This scaling ensures:

- Numerical stability
- Consistent dynamic range across inputs

2) *Train-Validation Split*: Let total samples be N :

$$N = |\mathcal{D}| \quad (42)$$

We partition:

$$N_{\text{train}} = \lfloor 0.8N \rfloor \quad (43)$$

$$N_{\text{val}} = N - N_{\text{train}} \quad (44)$$

Thus:

$$\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}}, \quad \mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{val}} = \emptyset \quad (45)$$

Important Design Justification:

- Validation set contains ground truth (GT)
- External test set (512×512) lacks GT and is computationally infeasible
- Therefore, validation split is used for objective computation

3) *Model Instantiation*: A model $\mathcal{M}_{\mathbf{x}}$ is constructed:

$$\mathcal{M}_{\mathbf{x}} = \text{HWViT}(\mathbf{x}) \quad (46)$$

Number of trainable parameters:

$$f_7(\mathbf{x}) = \sum_{p \in \mathcal{M}} \mathbb{I}(p \text{ trainable}) \cdot |p| \quad (47)$$

4) *Training Objective*: The model is trained using L1 loss:

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^B \|\hat{\mathbf{Y}}_i - \mathbf{Y}_i\|_1 \quad (48)$$

where:

- $\hat{\mathbf{Y}}$ is predicted output
- $\mathbf{Y}$ is ground truth
- B is batch size

5) *Optimization Procedure*: We use Adam optimizer:

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} \quad (49)$$

where:

- $\eta = x_5$ (learning rate)
- $\hat{m}_t, \hat{v}_t$ are bias-corrected moments

6) *Epoch-Based Training*: Training proceeds for:

$$E = \text{epochs} \quad (50)$$

Each epoch processes:

$$\left\lceil \frac{N_{\text{train}}}{B} \right\rceil \text{ batches} \quad (51)$$

Total gradient updates per evaluation:

$$U = E \times \left\lceil \frac{N_{\text{train}}}{B} \right\rceil \quad (52)$$

D. Validation-Based Objective Computation

After training, predictions are generated:

$$\hat{\mathcal{D}}_{\text{val}} = \mathcal{M}_{\mathbf{x}}(\mathcal{D}_{\text{val}}) \quad (53)$$

Metrics are computed:

$$f_1 = \frac{\text{SAM}}{10} \quad (54)$$

$$f_2 = \frac{\text{ERGAS}}{10} \quad (55)$$

$$f_3 = -\text{CC} \quad (56)$$

$$f_4 = \frac{\text{SD}}{10} \quad (57)$$

$$f_5 = -\frac{\text{SF}}{10} \quad (58)$$

$$f_6 = -\text{SSIM} \quad (59)$$

$$f_7 = \frac{\text{n_params}}{10^6} \quad (60)$$

E. NSGA-II Evolution Process

1) *Population and Generations*: Let:

$$N = \text{population size} \quad (61)$$

$$G = \text{number of generations} \quad (62)$$

Total evaluations:

$$N_{\text{eval}} = N \times G \quad (63)$$

Each evaluation involves a full training process, making this an extremely expensive optimization.

2) *Fitness Assignment*: Each individual receives:

- Pareto rank r
- Crowding distance d

Selection criterion:

$$\mathbf{x}_1 \prec \mathbf{x}_2 \text{ if } \begin{cases} r_1 < r_2 \\ \text{or} \\ r_1 = r_2 \text{ and } d_1 > d_2 \end{cases} \quad (64)$$

3) *Checkpointing*: At generation t , algorithm state $\mathcal{A}_t$ is saved:

$$\mathcal{A}_t = \{P_t, F_t, t\} \quad (65)$$

Upon restart:

$$t \leftarrow \text{algorithm.n_gen} \quad (66)$$

Progress resumes without loss of computation.

F. Computational Complexity

Each evaluation cost:

$$\mathcal{O}(E \cdot N_{\text{train}} \cdot C_{\text{model}}) \quad (67)$$

Total optimization cost:

$$\mathcal{O}(N \cdot G \cdot E \cdot N_{\text{train}} \cdot C_{\text{model}}) \quad (68)$$

where C_{model} is forward-backward pass complexity.

G. Key Design Insights

- **Nested optimization**: NSGA-II operates over a training loop
- **Validation-driven objectives**: avoids test leakage
- **Projection ensures feasibility**: critical for transformer architectures
- **Multi-objective trade-offs**: quality vs complexity
- **Checkpointing**: essential for long-running experiments

H. Summary

This formulation transforms hyperparameter tuning into a principled multi-objective optimization problem, where each candidate solution is rigorously evaluated through a full training-validation pipeline, enabling discovery of Pareto-optimal fusion architectures.

III. NSGA-II BASED MULTI-OBJECTIVE OPTIMIZATION FRAMEWORK

A. Overview

We formulate hyperspectral image fusion as a **computationally expensive multi-objective optimization problem**, where each candidate solution corresponds to a full training and validation cycle of a deep neural network.

Let the optimization problem be defined as:

$$\min_{\mathbf{x} \in \Omega} \mathbf{F}(\mathbf{x}) \quad (69)$$

where:

- $\mathbf{x}$ represents model hyperparameters
 - $\mathbf{F}(\mathbf{x})$ is obtained via a training-evaluation pipeline
- Unlike analytical objectives, $\mathbf{F}$ is computed through:

$$\mathbf{F}(\mathbf{x}) = \text{train_and_evaluate}(\mathbf{x}) \quad (70)$$

B. Search Space Definition

Each candidate solution is:

$$\mathbf{x} = [x_1, x_2, x_3, x_4, x_5] \quad (71)$$

$$x_1 \in [16, 64] \quad \text{PAN target channels} \quad (72)$$

$$x_2 \in [16, 64] \quad \text{MS target channels} \quad (73)$$

$$x_3 \in [4, 16] \quad \text{Head channel size} \quad (74)$$

$$x_4 \in [0.01, 0.3] \quad \text{Dropout rate} \quad (75)$$

$$x_5 \in [10^{-4}, 10^{-3}] \quad \text{Learning rate} \quad (76)$$

1) *Architectural Constraint Projection*: To ensure compatibility with multi-head attention:

$$C \bmod C_h = 0 \quad (77)$$

we project:

$$C^* = \max \left(C_h, \left\lfloor \frac{C}{C_h} \right\rfloor C_h \right) \quad (78)$$

This guarantees valid tensor reshaping:

$$\mathbb{R}^{B \times C \times H \times W} \rightarrow \mathbb{R}^{B \times H \times C_h \times H \times W} \quad (79)$$

C. Training Pipeline as Objective Function

1) *Dataset Normalization*: Given input tensors:

$$\mathbf{I} \in \{\text{PAN, MS, LMS, GT}\} \quad (80)$$

normalization is applied:

$$\tilde{\mathbf{I}} = \frac{\mathbf{I}}{r}, \quad r = 2047 \quad (81)$$

2) *Train-Validation Split*: Let total samples be N :

$$N_{train} = \lfloor 0.8N \rfloor \quad (82)$$

$$N_{val} = N - N_{train} \quad (83)$$

$$\mathcal{D} = \mathcal{D}_{train} \cup \mathcal{D}_{val} \quad (84)$$

This split is crucial because:

- Validation set contains ground truth
- External test set lacks GT and is computationally prohibitive

3) *Model Training*: The model $\mathcal{M}_{\mathbf{x}}$ is trained using L1 loss:

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^B \|\hat{Y}_i - Y_i\|_1 \quad (85)$$

Optimization uses Adam:

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (86)$$

4) *Training Budget Per Candidate*: Each candidate is trained for:

$$E = 5 \text{ epochs} \quad (87)$$

Let batch size be $B = 8$. Then:

$$\text{Batches per epoch} = \left\lceil \frac{N_{train}}{8} \right\rceil \quad (88)$$

Total updates per candidate:

$$U = 5 \times \left\lceil \frac{N_{train}}{8} \right\rceil \quad (89)$$

Interpretation:

- Small E reduces computational cost
- Acts as a proxy for final performance
- Enables large-scale exploration via NSGA-II

D. Objective Functions

The optimization involves 7 objectives:

$$f_1 = \frac{\text{SAM}}{10} \quad (90)$$

$$f_2 = \frac{\text{ERGAS}}{10} \quad (91)$$

$$f_3 = -\text{CC} \quad (92)$$

$$f_4 = \frac{\text{SD}}{10} \quad (93)$$

$$f_5 = -\frac{\text{SF}}{10} \quad (94)$$

$$f_6 = -\text{SSIM} \quad (95)$$

$$f_7 = \frac{n_{\text{params}}}{10^6} \quad (96)$$

All objectives are converted into minimization form.

E. NSGA-II Configuration

1) *Search Budget*:

$$N = 25 \quad (\text{population size}) \quad (97)$$

$$G = 20 \quad (\text{generations}) \quad (98)$$

Total number of candidate evaluations:

$$N_{\text{eval}} = N \times G = 25 \times 20 = 500 \quad (99)$$

2) *Total Training Cost*: Each evaluation requires $E = 5$ epochs:

$$\text{Total training epochs} = 500 \times 5 = 2500 \quad (100)$$

Total gradient updates:

$$\mathcal{O}(2500 \cdot N_{train}) \quad (101)$$

F. Evolution Process

At generation t :

$$P_t \rightarrow Q_t \rightarrow R_t = P_t \cup Q_t \quad (102)$$

Selection is based on:

- Pareto rank
- Crowding distance

Dominance condition:

$$\mathbf{x}_1 \prec \mathbf{x}_2 \iff \forall i : f_i(\mathbf{x}_1) \leq f_i(\mathbf{x}_2) \wedge \exists j : f_j(\mathbf{x}_1) < f_j(\mathbf{x}_2) \quad (103)$$

G. Model Persistence and Hash Mapping

Each candidate is uniquely identified:

$$h(\mathbf{x}) = \text{MD5}(\mathbf{x}) \quad (104)$$

Model weights:

$$\mathcal{M}_{\mathbf{x}} \rightarrow \text{model}_{h(\mathbf{x}).pt} \quad (105)$$

1) *Pareto Model Extraction*: For Pareto set $\mathcal{P}$:

$$\mathcal{P} = \{(\mathbf{x}_i, \mathbf{F}_i, \mathcal{M}_i)\} \quad (106)$$

Mapping stored as:

$$\text{decision}_i \leftrightarrow \text{result}_i \leftrightarrow \text{model}_i \quad (107)$$

2) *Storage Optimization*: Non-Pareto models are discarded:

$$\mathcal{M}_{\text{non-Pareto}} \rightarrow \emptyset \quad (108)$$

This reduces disk usage from:

$$\mathcal{O}(500) \rightarrow \mathcal{O}(|\mathcal{P}|) \quad (109)$$

H. Final Pareto Front Representation

The Pareto front is stored as:

$$\mathcal{P} = \{(\mathbf{x}_i, \mathbf{F}_i, \text{path}_i)\}_{i=1}^{|\mathcal{P}|} \quad (110)$$

with:

- Decision variables
- Objective values
- Model file paths

I. Key Insights

- **Nested optimization**: NSGA-II operates over full training loops
- **Compute-aware design**: small epochs enable tractability
- **Massive search**: 500 architectures explored
- **Efficient storage**: only Pareto-optimal models retained
- **Reproducibility**: deterministic hashing ensures traceability

J. Conclusion

The proposed framework integrates evolutionary optimization with deep learning training, enabling systematic exploration of architectural trade-offs under multiple conflicting objectives. By combining NSGA-II with a carefully designed training pipeline and efficient model management, the method produces a diverse and computationally tractable Pareto front of high-quality fusion models.

IV. HIERARCHICAL WAVELET VISION TRANSFORMER (HWVIT) ARCHITECTURE

A. Overview

The proposed **HWVIT (Hierarchical Wavelet Vision Transformer)** is a multi-scale fusion architecture designed for hyperspectral image fusion. It integrates:

- **Wavelet decomposition** for frequency separation
- **Transformer-based attention** for global context modeling
- **Multi-scale hierarchical fusion** for spatial consistency
- **CNN-based refinement** for local detail reconstruction

The model takes as input:

$$\text{PAN} \in \mathbb{R}^{B \times C_p \times H \times W}, \quad \text{MS} \in \mathbb{R}^{B \times C_m \times \frac{H}{4} \times \frac{W}{4}}, \quad \text{LMS} \in \mathbb{R}^{B \times C_m \times H \times W} \quad (111)$$

and produces:

$$\hat{Y} \in \mathbb{R}^{B \times C_m \times H \times W} \quad (112)$$

B. Wavelet Transform (DWT and IDWT)

1) *Discrete Wavelet Transform*: The DWT decomposes an image into four frequency subbands:

$$\{LL, LH, HL, HH\} \quad (113)$$

Each is computed via convolution:

$$LL = x * w_{LL}, \quad LH = x * w_{LH} \quad (114)$$

$$HL = x * w_{HL}, \quad HH = x * w_{HH} \quad (115)$$

where filters are:

$$w_{LL} = \begin{bmatrix} 0.25 & 0.25 \\ 0.25 & 0.25 \end{bmatrix} \quad w_{LH}, w_{HL}, w_{HH} \text{ similarly defined} \quad (116)$$

Downsampling is performed with stride 2:

$$x_{LL} \in \mathbb{R}^{B \times C \times \frac{H}{2} \times \frac{W}{2}} \quad (117)$$

2) *Inverse Wavelet Transform*: Reconstruction is:

$$x = LL \uparrow + LH \uparrow + HL \uparrow + HH \uparrow \quad (118)$$

implemented via transposed convolution:

$$x = \sum_{i \in \{LL, LH, HL, HH\}} x_i * w_i^T \quad (119)$$

C. Channel Transformation Modules

1) Channel Expansion (raise_channel):

$$x' = \sigma(W_2 * \sigma(W_1 * x)) \quad (120)$$

where:

- W_1 : 5×5 convolution
- W_2 : 3×3 convolution

Purpose:

- Increase representational capacity
- Align feature dimensionality

2) Channel Reduction:

$$x' = W_3 * \sigma(W_2 * \sigma(W_1 * x)) \quad (121)$$

Used to project features back to original MS dimensionality.

D. Feed Forward Networks

1) Transformer FFN:

$$\text{FFN}(x) = W_2 \cdot \sigma(W_1 x) \quad (122)$$

with spatial reshaping:

$$x \in \mathbb{R}^{B \times C \times H \times W} \rightarrow \mathbb{R}^{B \times HW \times C} \quad (123)$$

2) Convolutional FFN:

$$x' = W_4 * \sigma(W_2 * (W_1 * x)) + W_1 * x \quad (124)$$

This introduces:

- Local receptive field
- Residual learning

E. Attention Mechanism

1) Multi-Head Attention: Given input x :

$$Q = W_Q x, \quad K = W_K x, \quad V = W_V x \quad (125)$$

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (126)$$

where:

$$d = \text{head_channel} \quad (127)$$

Multi-head formulation:

$$\text{MHA}(x) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \quad (128)$$

2) Residual MLP Enhancement:

$$x_1 = x + \text{MLP}_1(\text{Attention}(x)) \quad (129)$$

$$x_2 = x_1 + \text{MLP}_2(x_1) \quad (130)$$

F. Dynamic Weighting (DWC)

Applies channel-wise modulation:

$$w = \sigma(Wx) \quad (131)$$

$$x' = w \odot x \quad (132)$$

This enables:

- Adaptive feature scaling
- Frequency-aware modulation

G. Residual Blocks

$$y = x + W_2 * \sigma(W_1 * x) \quad (133)$$

Used extensively for:

- Stability
- Gradient flow

H. Wavelet-Transformer Blocks

1) S_MWT (Single-scale): Pipeline:

1) Apply DWT:

$$x \rightarrow \{LL, LH, HL, HH\} \quad (134)$$

2) Project to MS space:

$$LL' = W_{proj} LL \quad (135)$$

3) Compute attention:

$$V_{LL} = \text{Attention}(LL', LL', V) \quad (136)$$

4) Reconstruct via IDWT:

$$x_{attn} = \text{IDWT}(V_{LL}, V_{LH}, V_{HL}, V_{HH}) \quad (137)$$

5) Combine with spatial path:

$$x = \text{Conv}(x_{attn}) + \text{Conv}(x_{wavelet}) \quad (138)$$

2) F_MWT (Full-scale): Combines:

$$x = \alpha \cdot \text{PAN} + \beta \cdot \text{LMS} + (1 - \alpha - \beta) \cdot \text{Transformer} \quad (139)$$

3) L_MWT (Low-scale): Operates on downsampled inputs:

$$x \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2}} \quad (140)$$

Captures:

- Global structure
- Low-frequency consistency

I. Hierarchical Pipeline

1) Step 1: Upsampling:

$$\text{MS} \xrightarrow{\text{PixelShuffle}} \text{LMS} \quad (141)$$

2) Step 2: Multi-Scale Fusion:

$$\text{Low-scale: } \text{back}_1 = L_MWT(\dots) \quad (142)$$

$$\text{Full-scale: } \text{back}_2 = F_MWT(\dots) \quad (143)$$

3) Step 3: Reconstruction:

$$\hat{Y} = \sigma(\text{Reduce}(\text{back}_2) + \text{LMS}) \quad (144)$$

J. Final Model Equation

The complete model can be summarized as:

$$\hat{Y} = \mathcal{F}_{HWWiT}(\text{PAN}, \text{MS}, \text{LMS}) \quad (145)$$

where:

$$\mathcal{F} = \text{Reduce} \circ F_MWiT \circ L_MWiT \circ \text{Upsample} \quad (146)$$

K. Key Design Insights

- **Wavelet decomposition:** separates frequency components explicitly
- **Attention:** models long-range dependencies
- **Hierarchical structure:** captures both global and local features
- **Residual learning:** stabilizes deep training
- **Multi-branch fusion:** balances spectral and spatial information

L. Computational Complexity

Let $N = HW$:

- Attention: $\mathcal{O}(N^2)$
- Wavelet: $\mathcal{O}(N)$
- Convolution: $\mathcal{O}(k^2 N)$

Overall:

$$\mathcal{O}(N^2) \quad (\text{dominated by attention}) \quad (147)$$

M. Conclusion

The HWWiT architecture combines wavelet-based frequency decomposition with transformer-based global reasoning in a hierarchical framework, enabling effective fusion of PAN and MS imagery while preserving both spatial detail and spectral fidelity.

V. EVALUATION METRICS

A. Overview

To comprehensively evaluate the quality of the fused hyperspectral images, we employ a set of **seven complementary metrics** capturing spectral fidelity, spatial detail preservation, statistical similarity, and perceptual quality.

Let:

$$\hat{\mathbf{Y}} \in \mathbb{R}^{B \times C \times H \times W}, \quad \mathbf{Y} \in \mathbb{R}^{B \times C \times H \times W} \quad (148)$$

denote the predicted and ground-truth hyperspectral images, respectively.

B. Spectral Angle Mapper (SAM)

1) *Definition:* SAM measures the angular difference between spectral vectors:

$$\text{SAM} = \frac{1}{N} \sum_{i=1}^N \arccos \left(\frac{\hat{\mathbf{y}}_i \cdot \mathbf{y}_i}{\|\hat{\mathbf{y}}_i\|_2 \cdot \|\mathbf{y}_i\|_2 + \epsilon} \right) \quad (149)$$

where:

- $\hat{\mathbf{y}}_i, \mathbf{y}_i \in \mathbb{R}^C$ are spectral vectors at pixel i
- $N = H \times W$
- ϵ prevents division by zero

2) *Interpretation:*

- Measures **spectral distortion**
- Invariant to illumination scaling
- Lower values indicate better spectral fidelity

C. Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS)

1) *Definition:*

$$\text{ERGAS} = \frac{100}{r} \sqrt{\frac{1}{C} \sum_{c=1}^C \left(\frac{\text{RMSE}_c}{\mu_c} \right)^2} \quad (150)$$

where:

$$\text{RMSE}_c = \sqrt{\frac{1}{HW} \sum_i (\hat{Y}_{c,i} - Y_{c,i})^2} \quad (151)$$

$$\mu_c = \frac{1}{HW} \sum_i Y_{c,i} \quad (152)$$

and r is the resolution ratio (typically $r = 4$).

2) *Interpretation:*

- Measures **global relative reconstruction error**
- Normalized across spectral bands
- Lower values indicate better reconstruction

D. Correlation Coefficient (CC)

1) *Definition:*

$$\text{CC} = \frac{\sum_i (\hat{Y}_i - \bar{\hat{Y}})(Y_i - \bar{Y})}{\sqrt{\sum_i (\hat{Y}_i - \bar{\hat{Y}})^2} \cdot \sqrt{\sum_i (Y_i - \bar{Y})^2}} \quad (153)$$

2) *Interpretation:*

- Measures **linear correlation** between prediction and ground truth
- Range: $[-1, 1]$
- Higher values indicate stronger similarity

E. Standard Deviation Difference (SD)

1) *Definition:*

$$\text{SD} = \left| \sigma(\hat{\mathbf{Y}}) - \sigma(\mathbf{Y}) \right| \quad (154)$$

where:

$$\sigma(X) = \sqrt{\frac{1}{N} \sum_i (X_i - \bar{X})^2} \quad (155)$$

2) Interpretation:

- Measures **contrast preservation**
- Reflects global intensity distribution similarity
- Lower values indicate better statistical consistency

F. Spatial Frequency (SF)

1) *Definition:* Spatial frequency measures spatial detail variation:

$$RF = \sqrt{\frac{1}{HW} \sum_{i,j} (I_{i,j+1} - I_{i,j})^2} \quad (156)$$

$$CF = \sqrt{\frac{1}{HW} \sum_{i,j} (I_{i+1,j} - I_{i,j})^2} \quad (157)$$

$$SF = \sqrt{RF^2 + CF^2} \quad (158)$$

2) Interpretation:

- Captures **spatial detail and texture richness**
- High values indicate sharper images
- Encourages preservation of high-frequency details

G. Structural Similarity Index (SSIM)

1) *Definition:* For each spectral band:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (159)$$

Final SSIM:

$$SSIM_{final} = \frac{1}{C} \sum_{c=1}^C SSIM_c \quad (160)$$

2) Interpretation:

- Measures **perceptual similarity**
- Combines luminance, contrast, and structure
- Range: $[0, 1]$
- Higher values indicate better visual quality

H. Edge Preservation (Conceptual)

Although not directly used in optimization, edge preservation can be defined as:

$$E = \frac{1}{N} \sum_i |\nabla I_{pan}(i)| \quad (161)$$

where ∇ denotes spatial gradient.

1) Interpretation:

- Measures edge strength
- Reflects spatial detail transfer from PAN

I. Summary of Metrics

J. Multi-Objective Role

For optimization, all metrics are converted into minimization objectives:

$$\mathbf{F} = \left[\frac{SAM}{10}, \frac{ERGAS}{10}, -CC, \frac{SD}{10}, -\frac{SF}{10}, -SSIM \right] \quad (162)$$

This ensures compatibility with NSGA-II.

Metric	Property	Goal
SAM	Spectral distortion	Minimize
ERGAS	Global error	Minimize
CC	Correlation	Maximize
SD	Contrast difference	Minimize
SF	Spatial detail	Maximize
SSIM	Perceptual quality	Maximize

K. Conclusion

The selected metrics jointly evaluate spectral accuracy, spatial fidelity, statistical consistency, and perceptual quality, providing a robust and comprehensive assessment of hyper-spectral image fusion performance.

VI. COMPARISON PIPELINE AND EVALUATION FRAMEWORK

A. Overview

This section describes the complete evaluation pipeline used to compare the proposed deep learning-based fusion method (GOD: Generated Optimal Design via NSGA-II) against classical pansharpening techniques, namely:

- Brovey Transform
- Intensity-Hue-Saturation (IHS) Transform

For each test sample and each Pareto-optimal model, fused outputs are generated and evaluated using a set of quantitative metrics.

B. Data Normalization

All input data is normalized using a fixed scaling factor:

$$X_{\text{norm}} = \frac{X}{\alpha}, \quad \alpha = 2047 \quad (163)$$

This ensures that pixel intensities lie approximately within the range $[0, 1]$, which stabilizes neural network inference and ensures consistency across evaluation methods.

C. Input Representation

Let:

- $\mathbf{P} \in \mathbb{R}^{H \times W \times 1}$ denote the panchromatic image
- $\mathbf{M} \in \mathbb{R}^{h \times w \times C}$ denote the low-resolution multispectral image
- $\mathbf{Y} \in \mathbb{R}^{H \times W \times C}$ denote the ground truth

The multispectral image is upsampled using bicubic interpolation:

$$\mathbf{M}_{\uparrow} = \text{Interp}_{\text{bicubic}}(\mathbf{M}, H, W) \quad (164)$$

D. Proposed Model (GOD Pipeline)

The trained HWViT model performs:

$$\hat{\mathbf{Y}} = f_{\theta}(\mathbf{P}, \mathbf{M}, \mathbf{M}_{\uparrow}) \quad (165)$$

where:

- θ are model parameters optimized via NSGA-II
- $\mathbf{M}_{\uparrow}$ provides spatial alignment

E. Brovey Transform

The Brovey method enhances spatial resolution using intensity modulation.

1) *Mathematical Formulation:* Let:

$$I(x, y) = \sum_{c=1}^C M_{\uparrow}^{(c)}(x, y) \quad (166)$$

Then the fused output is:

$$\hat{Y}^{(c)}(x, y) = \frac{M_{\uparrow}^{(c)}(x, y)}{I(x, y) + \epsilon} \cdot P(x, y) \quad (167)$$

2) *Interpretation:*

- Preserves spectral ratios of MS image
- Injects spatial detail from PAN
- Sensitive to spectral distortion

F. IHS Transform

The IHS method separates intensity from chromatic components.

1) *Forward Transformation:* Given RGB channels:

$$I = \frac{R + G + B}{3} \quad (168)$$

$$S = 1 - \frac{\min(R, G, B)}{I + \epsilon} \quad (169)$$

Hue is computed as:

$$\theta = \arccos \left(\frac{0.5[(R - G) + (R - B)]}{\sqrt{(R - G)^2 + (R - B)(G - B)} + \epsilon} \right) \quad (170)$$

$$H = \begin{cases} \theta & B \leq G \\ 2\pi - \theta & B > G \end{cases} \quad (171)$$

2) *Fusion Step:* The intensity component is replaced:

$$I' = P \quad (172)$$

3) *Inverse Transformation:* RGB channels are reconstructed using piecewise transformations depending on hue regions:

$$(R, G, B) = \mathcal{T}^{-1}(I', H, S) \quad (173)$$

4) *Interpretation:*

- Separates spatial and spectral information
- Can introduce spectral distortion due to intensity substitution

G. Metric Computation

For each predicted image $\hat{Y}$, metrics are computed:

$$\mathbf{m} = [\text{SAM}, \text{ERGAS}, \text{CC}, \text{SD}, \text{SF}, \text{SSIM}] \quad (174)$$

H. Objective Restoration

Since NSGA-II converts all objectives into minimization form:

$$\mathbf{F} = \left[\frac{\text{SAM}}{10}, \frac{\text{ERGAS}}{10}, -\text{CC}, \frac{\text{SD}}{10}, -\frac{\text{SF}}{10}, -\text{SSIM} \right] \quad (175)$$

The original values are restored as:

$$\text{SAM} = 10 \cdot F_1, \quad \text{ERGAS} = 10 \cdot F_2 \quad (176)$$

$$\text{CC} = -F_3, \quad \text{SF} = -10 \cdot F_5, \quad \text{SSIM} = -F_6 \quad (177)$$

I. Min-Max Normalization

To compare models across different metric scales, normalization is applied:

$$\tilde{x}_i = \frac{x_i - \min(x)}{\max(x) - \min(x) + \epsilon} \quad (178)$$

1) *Purpose:*

- Converts all metrics to $[0, 1]$ scale
- Enables fair visual comparison
- Removes unit and scale dependency

J. Comparative Visualization

For each test index:

- GOD models are plotted as bar graphs
- Baseline methods (Brovey/IHS) are plotted as horizontal reference lines

1) *Interpretation:*

- Higher normalized values indicate relative performance within the set
- Visualization highlights Pareto dominance

K. Data Storage and Organization

For each model and test index:

- Input images (PAN, MS)
- Ground truth
- Predicted output
- Metric logs

are stored systematically for reproducibility.

L. Pipeline Summary

The complete pipeline can be summarized as:

- 1) Load dataset and normalize inputs
- 2) Upsample multispectral image
- 3) Generate predictions:
 - HWViT (GOD models)
 - Brovey transform
 - IHS transform
- 4) Compute evaluation metrics
- 5) Restore original metric values
- 6) Apply min-max normalization
- 7) Generate comparative plots

M. Conclusion

This evaluation framework ensures:

- Fair comparison across classical and deep models
- Multi-dimensional performance assessment
- Clear visualization of trade-offs among Pareto-optimal solutions

VII. PARETO FRONT VISUALIZATION PIPELINE

A. Overview

This section describes the visualization framework used to analyze the Pareto-optimal solutions obtained from NSGA-II. The goal is to interpret high-dimensional objective trade-offs using:

- Parallel Coordinate Plots (PCP)
- 2D Pareto projections
- 3D Pareto projections

Let the Pareto objective matrix be:

$$\mathbf{F} \in \mathbb{R}^{N \times M} \quad (179)$$

where:

- N = number of Pareto-optimal solutions
- M = number of objectives (here $M = 7$)

Each row corresponds to a solution:

$$\mathbf{f}_i = [f_{i1}, f_{i2}, \dots, f_{iM}] \quad (180)$$

B. Objective Representation

The objectives are:

$$\text{OBJ_NAMES} = [\text{SAM}, \text{ERGAS}, \text{CC}, \text{SD}, \text{SF}, \text{SSIM}, n_{\text{params}}] \quad (181)$$

However, during NSGA-II optimization, maximization objectives are converted into minimization form:

$$\mathbf{F} = \left[\frac{\text{SAM}}{10}, \frac{\text{ERGAS}}{10}, -\text{CC}, \frac{\text{SD}}{10}, -\frac{\text{SF}}{10}, -\text{SSIM}, \frac{n_{\text{params}}}{10^6} \right] \quad (182)$$

C. Objective Restoration

To obtain human-interpretable values, negated objectives are restored:

$$f_{ij}^{\text{display}} = \begin{cases} -f_{ij}, & j \in \{2, 4, 5\} \\ f_{ij}, & \text{otherwise} \end{cases} \quad (183)$$

This ensures:

- CC, SF, SSIM regain their original meaning (higher is better)
- All metrics are interpretable in their natural domain

D. Parallel Coordinate Plot (PCP)

1) *Normalization*: Each objective is normalized independently:

$$\tilde{f}_{ij} = \frac{f_{ij} - \min_i f_{ij}}{\max_i f_{ij} - \min_i f_{ij} + \epsilon} \quad (184)$$

This ensures:

- All objectives lie in $[0, 1]$
- Different scales (e.g., SAM vs n_{params}) become comparable

2) *Structure*: In PCP:

- Each vertical axis represents one objective
- Each solution is plotted as a polyline:

$$\tilde{\mathbf{f}}_i = [\tilde{f}_{i1}, \dots, \tilde{f}_{iM}] \quad (185)$$

3) *Interpretation*:

- Intersections indicate trade-offs between objectives
- Parallel lines indicate similar performance
- Outliers indicate extreme solutions

E. 2D Pareto Projections

For each pair of objectives (j, k) :

$$\mathcal{P}_{jk} = \{(f_{ij}, f_{ik}) \mid i = 1, \dots, N\} \quad (186)$$

1) *Purpose*:

- Visualize trade-offs between two objectives
- Identify dominance relationships
- Detect convexity or clustering in the Pareto front

2) *Example*:

- SAM vs SSIM $\rightarrow$ spectral vs perceptual trade-off
- ERGAS vs n_{params} $\rightarrow$ accuracy vs complexity

F. 3D Pareto Projections

For triplets (j, k, l) :

$$\mathcal{P}_{jkl} = \{(f_{ij}, f_{ik}, f_{il}) \mid i = 1, \dots, N\} \quad (187)$$

1) *Purpose*:

- Capture interactions among three objectives
- Reveal hidden trade-off structures not visible in 2D

G. Combinatorial Coverage

All possible projections are generated:

1) *2D Combinations*:

$$\binom{M}{2} = \frac{M(M-1)}{2} \quad (188)$$

For $M = 7$:

$$\binom{7}{2} = 21 \quad (189)$$

2) *3D Combinations:*

$$\binom{M}{3} = \frac{M(M-1)(M-2)}{6} \quad (190)$$

For $M = 7$:

$$\binom{7}{3} = 35 \quad (191)$$

H. Visualization Workflow

1) Load Pareto results:

$$\mathbf{F} \leftarrow \text{pickle file} \quad (192)$$

2) Restore objective values:

$$\mathbf{F}_{\text{display}} = \mathcal{R}(\mathbf{F}) \quad (193)$$

3) Generate PCP:

- Normalize each objective
- Plot all solutions

4) Generate all 2D projections:

- Scatter plots for each objective pair

5) Generate all 3D projections:

- 3D scatter plots for each objective triplet

I. Computational Considerations

- PCP complexity: $\mathcal{O}(NM)$
- 2D plots: $\mathcal{O}(N \cdot \binom{M}{2})$
- 3D plots: $\mathcal{O}(N \cdot \binom{M}{3})$

J. Interpretation of Pareto Front

The Pareto front represents solutions satisfying:

$$\nexists \mathbf{f}_j \text{ such that } f_{j,k} \leq f_{i,k} \forall k \text{ and } f_{j,k} < f_{i,k} \text{ for at least one } k \quad (194)$$

1) *Insights:*

- No single solution is globally optimal
- Trade-offs must be balanced
- Visualization aids decision-making

K. Conclusion

This visualization pipeline transforms high-dimensional optimization results into interpretable graphical representations, enabling:

- Analysis of trade-offs across objectives
- Identification of optimal operating points
- Insight into model complexity vs performance